

# VERNIER: an open-source library pushing marker pose estimation to nanometric resolutions

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 17 March 2026

Published: unpublished

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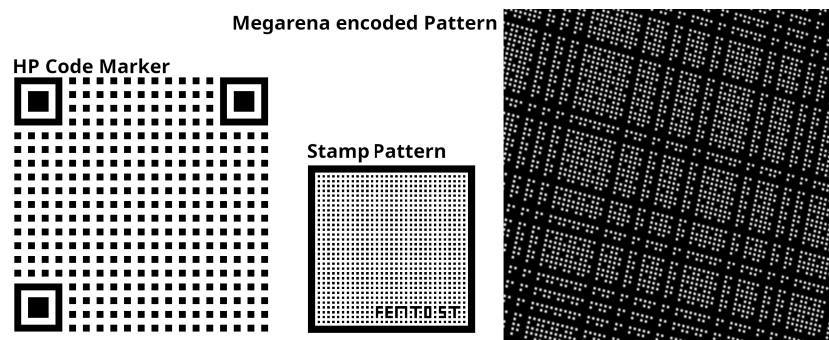
## Summary

At the small scales, few solutions exist to capture the pose of an object with nanometric and microradian resolutions over relatively large ranges. Over the years, we have proposed several fiducial marker and pattern designs to achieve reliable performance for various microscopy applications. VERNIER is an open source phase processing library designed to provide accurate pose measurement based on these results. Centimeter ranges are possible using binary encoding methods, while nanometer resolutions can be achieved using phase processing of the periodic patterns. Moreover, the detection has proven to be particularly robust to noise, defocus and occlusion thanks to a local thresholding algorithm based on the phase. The paper presents the measurement principles and the patterns implemented in VERNIER addressing different application needs.

## Statement of need

The VERNIER library provides a powerful tool to measure the 2D and 3D poses of objects observed through a microscope lens with unprecedented sub-pixel resolutions and ranges. The library works in combination with markers fixed on the objects of interest, allowing accurate measurements even in noisy and blurred imaging contexts which commonly occurs in the microscopy context. The image processing involves two complementary steps, resulting on the one hand in fine but relative position measurement and, on the other hand, a coarser but absolute localization. The fine position process achieves high sub-pixel resolutions thanks to spectral analysis of the imaged periodic pattern of the marker. Complementary coarse measurements are retrieved either from the marker contours or from a binary sequence encrypted within the pattern.

VERNIER works with three different types of markers, designed to meet different application requirements, and whose design can be easily adapted to specific imaging parameters, mainly magnification, field of view (FoV), and expected measurement range and resolution (cf. [Figure 1](#)). Sources and layouts are available on [GitHub](#) and can be downloaded for evaluation and testing.



**Figure 1:** The three marker types (on the left an HP code marker, in the middle a Stamp marker, on the right a Megarena encoded pattern).

## State of the field

Computer vision is widely used to track the movement of people and objects in many applications. When the size of the object of interest decreases, pose estimation becomes challenging due to the constraint of microscopy imaging. Unlike regular cameras, microscopes suffer from narrow FoV, short depths of field, low contrasts and out-of-focus blurs, and usual fiducial markers perform poorly in these conditions.

Many vision-based methods have been proposed to tackle pose estimation at the small scales. In 2021, Fatikow published a review paper comparing the resolution and the range of state-of-the-art vision-based localization methods (Yao et al., 2021). The most precise methods use phase correlation and can achieve sub-nanometer resolutions. However, their measurement ranges are still limited by the microscope's FoV. To overcome this limitation, pseudo-periodic patterns can be used to encode the absolute position over centimetric ranges, while using phase measurement to achieve nanometer resolutions (André et al., 2020b, 2020a). Based on this principle, VERNIER proposed processing algorithms for several markers and patterns that ensure reliable performance in various microscopy applications. This approach outperforms all others in terms of range-to-resolution ratio (Yao et al., 2021).

The measurement principle is mainly suited for in-plane 3 degrees of freedom (DoF) pose estimation under microscopy orthographic projection. However, long-focal perspective projection can be used for retrieving complementary out-of-plane pose parameters with a lower resolution. Full out-of-plane pose estimation details and performances can be found in (André, Sandoz, et al., 2022).

To achieve the measurement of the 6 DoF, the same approach is applied but with a digital holographic microscope (DHM). The interferometric character of DHM makes the device highly sensitive to out-of-plane motion and the 6 DoF are measured simultaneously in a single image with a high resolution (Ahmad et al., 2024).

Figure 2 presents the mean features and metrics of the pose estimation of all marker types. All the metrics have been validated experimentally with precision stages and robots (Ahmad et al., 2024; André et al., 2020b, 2020a; André, Sandoz, et al., 2022).

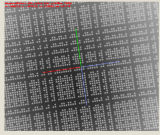
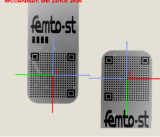
|                                                                                                                  | Optical microscopy<br>(orthographic projection)                   | Digital holographic<br>microscopy<br>(orthographic projection<br>+ interferometry)                                                                         | Long –focus lens<br>(perspective projection<br>without distortion)                                                                                            |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute pose<br>estimation<br> | 2D (x, y, rz)<br>Resolutions: 1nm, 4μrad<br>Ranges: 110mm, 2π rad | 3D (x, y, z, rx, ry, rz)<br>In-plane res: 1nm, 4μrad<br>In-plane ranges: 110mm, 2π<br>Out-of-plane res: 1nm, 0.1μrad<br>Out-of-plane ranges: 0.2mm, 0.2rad | 3D (x, y, z, rx, ry, rz)<br>In-plane res: 0.1μm, 4μrad<br>In-plane ranges: 150mm, 2π<br>Out-of-plane res: 0.1mm, 0.1mrad<br>Out-of-plane ranges: 10mm, 0.4rad |
| Relative pose<br>estimation<br> | 2D (x, y, rz)<br>Resolutions: 5nm, 10μrad<br>Ranges: fov, 2π rad  | 3D (x, y, z, rx, ry, rz)<br>Resolutions: not evaluated yet<br>In-plane ranges: fov, 2π rad<br>Out-of-plane ranges: 0.2mm, 0.2rad                           | 3D (x, y, z, rx, ry, rz)<br>Resolutions: not evaluated yet<br>In-plane ranges: fov, 2π rad<br>Out-of-plane ranges: dof, 0.4rad                                |

Figure 2: Overview of features and metrics of the marker pose estimation with VERNIER.

62 **Software design**

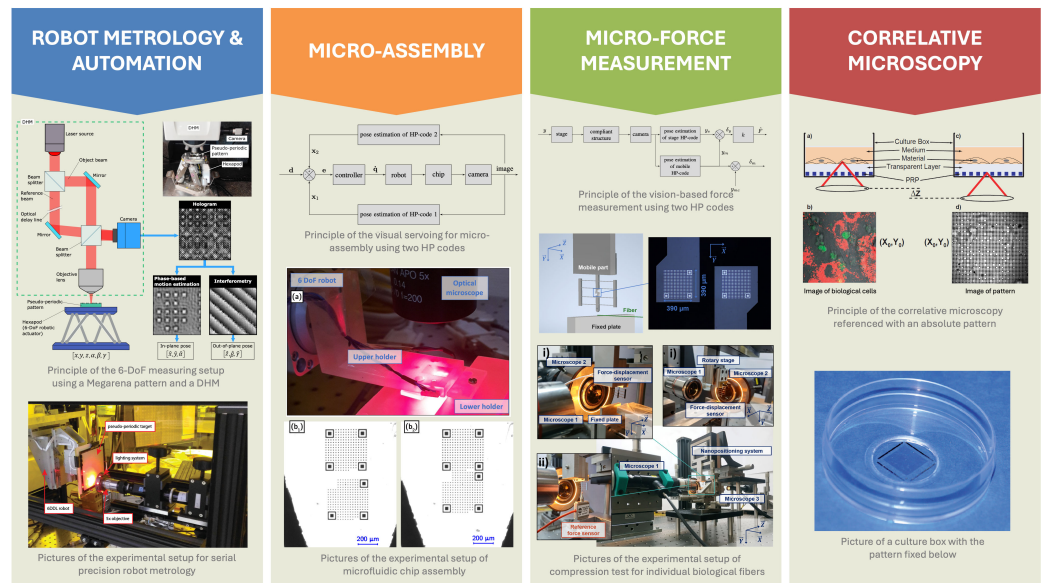
63 VERNIER is written in C++ and defines a collection of classes for detection and estimating the  
64 pose of three kinds of calibrated patterns (Megarena patterns, HP code and Stamp markers).  
65 The library is cross-platform. The users can build the examples and unit tests using CMake.  
66 The library relies on multiple third parties: OpenCV, Eigen, FFTW, MatIO, RapidJSON and GDS  
67 Tool Kit.

68 The library also provides a set of classes for rendering synthetic images and exporting layouts  
69 of the different markers presented in previous section. The synthetic images are used to test  
70 the correct functioning of the detectors. The marker layouts can be exported in PNG, SVG, GDS  
71 and OASIS formats. These files can be used to print the markers on various supports.

72 As the measurement method is based on the phase of the periodic pattern, the position  
73 accuracy is directly related to the scale and quality of the pattern fabrication. To obtain  
74 calibrated measures, all the patterns have been realized using a high-resolution maskless aligner  
75 (Heidelberg MLA150) on quartz or glass substrates within the [MIMENTO facility](#).

76 **Research impact statement**

77 VERNIER has found many applications in microrobotics as presented in [Figure 3](#).



**Figure 3:** Overview of applications of pose estimation at the small scales with VERNIER.

78 Metrology of precision manipulators One of the major interest of Megarena patterns is to  
79 perform the metrology of micro and nano stages and precision manipulators. Indeed, few  
80 solutions to measure the 3D pose of the manipulator end-effector at the nanoscale are available.  
81 The laser interferometers provide a very high range-to-resolution ratio of approximately  $10^9$   
82 but only along the laser's axis. Setups with several interferometers have demonstrated multiple  
83 DoF measurement systems [Lee et al. (2011); ortlepp2024high], at the expense of occupied  
84 volume and calibration complexity. Moreover, due to the constraints of laser reflection, the  
85 range of their angular measurements is very low, not exceeding a magnitude of one milliradian.

86 The use of Megarena patterns is much simpler and also provides nanometric resolution and  
87 centimeter ranges. Moreover, using a DHM, the angular ranges of out-of-plane rotations reach  
88 0.2 rad with resolutions down to  $0.1 \mu\text{rad}$ . The ideal setup consists only of a microscope and a  
89 piece of a Megarena pattern attached to the end effector of the manipulator.

90 For example, the method has been used to evaluate the accuracy of a precision hexapod,  
91 showing that it is able to follow millimeter trajectories with deviations of less than one  
92 micrometer (Ahmad et al., 2024). It has also been employed to evaluate the accuracy of a  
93 serial 6-DoF manipulator over a point matrix of 10 by 10 centimeters (André et al., 2020b).  
94 Megarena patterns can also be used to measure the repeatability of positioning of stages and  
95 manipulators. For instance, in (Gallardo et al., 2021; Mauzé et al., 2020), the method is used  
96 to evaluate the precision of parallel continuum robots.

97 Beyond stages and manipulators metrology, the method could also be useful for measuring the  
98 deviation of CNC machines and vision measuring machines.

99 Multi-DoF stage automation The Megarena patterns can also be used as position sensor  
100 to directly control the position of the motion platform of a multi-DoF manipulator during  
101 its operation. This ensures the actual position of the end-effector regardless of the errors  
102 introduced by assembly of stages, guidance, compliance, and backlash in the mechanical axis.  
103 It could also help to identify and correct cross-axis coupling and unwanted motion in multi-DoF  
104 systems, as proposed in (Tan et al., 2015).

105 As no internal additional sensors are required, pattern-based direct measurement can be applied  
106 to an existing manipulator to improve its accuracy and to carry on precision tasks such as  
107 micro-assembly.

108 Micro-assembly Micro-assembly requires two parts to be positioned relatively to each other  
109 with high precision. To make assemblies with errors lower than a micrometer, the joint sensors  
110 of the manipulators are not sufficient and it is necessary to use visual servoing to control the  
111 relative position of the parts. To that extent, markers are placed on each of the parts to  
112 assemble and their relative position is monitored in real time during the assembly process. In  
113 (André, Lehmann, et al., 2022), two HP code markers are used to align and bond two parts of  
114 a microfluidic chip. This automated assembly achieved a positioning accuracy of better than  
115 50 nm.

116 Beyond this assembly example, HP code and Stamp markers could also be useful for wafer  
117 alignment in photolithography processes, die assembly in the semiconductor industry, and  
118 stitching in high-resolution 3D printing and laser processing.

119 Micro-force measurement Another key application in microrobotics is simultaneous force  
120 and displacement sensing. Force sensing has always been an issue for micro- and nanoscale  
121 applications. Force ranges from a few mN to a few hundred mN are typically required for  
122 manipulation, assembly and characterization tasks in most fields.

123 In (André, Lehmann, et al., 2022), two HP code markers have been attached to a compliant  
124 mechanism. The relative motion between the markers gives an image of the compression  
125 force, while the relative position of the markers to the camera frame gives the displacement of  
126 the mechanism base. This force sensor has been used experimentally to perform automated  
127 compression tests on individual synthetic and natural fibers, providing very rich information  
128 and allowing to better understand the mechanical fiber behavior (Govilas et al., 2024).

129 Previous experiments with this method have shown resolution measurements down to 50 nN  
130 with experimental ranges of 50 mN (Guelpa et al., 2015). The same principle has also been  
131 applied to multiple directions force sensing in (Tiwari et al., 2021).

132 Correlative Microscopy The pattern-based position measurement was initially developed to  
133 perform the repositioning of Petri dishes under a microscope for live cell monitoring (Galeano  
134 Z et al., 2011). Thanks to a pattern similar in principle to the Megarena, regions of interest  
135 of a culture dish were easily retrieved after transfers from a cell incubator to the microscope  
136 stage. Thanks to this approach, images of single cells can be captured at different time steps  
137 in a repeatable manner, as the cells can be identified based on their absolute position in the  
138 Petri dish.

139 In this way, Megarena patterns could also be used in correlative microscopy, to retrieve the  
140 position of the very same cell or tissue area and image it with various modalities (e.g. electron,  
141 confocal, fluorescent microscopes).

## 142 AI usage disclosure

143 No generative AI tools were used in the development of this software, the writing of this  
144 manuscript, or the preparation of supporting materials.

## 145 Acknowledgements

146 This work has been supported by the ANR project Holo-Control (ANR-21-CE42-0009) and by  
147 the Bourgogne-Franche-Comté Region (Nano6D). It has been achieved in the frame of the  
148 EIPHI Graduate School (ANR-17-EURE-0002). The encoded targets were realized thanks to the  
149 RENATECH technological network and its FEMTO-ST facility MIMENTO. The experiments  
150 were conducted within the ROBOTEX network (ANR-21-ESRE-0015) and its FEMTO-ST  
151 technological facility CMNR.

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